

Blip_OS — manual

Version: 2026-09-18. Without SHIFT, P1 controls internal clock speed and rungler influence; with SHIFT, it controls oscillator A rate. Hold DICE for about 1.8 seconds to change Run/Touch mode. TS opens the shared synthesis core.

BlipOS is the cross-coupled, clocked-chaos operating system for diceFriend. It is inspired by the Blippoo Box approach: two triangle oscillators drive interacting shift registers, sample-and-hold feedback, and a pair of resonant peak channels. The result moves between unstable bass pulses, ringing filters, interlocking patterns, gritty drones, and sharp clocked strikes.

BlipOS is especially useful for self-generating electronic textures, irregular rhythmic structures, resonant noise, and sounds that continuously reorganize themselves without losing their underlying pulse.

BlipOS has two performance modes: **Run mode** and **Touch mode**. Run mode keeps the cross-coupled system continuously active. Touch mode gates the same synthesis core from the touch pads, CV, or MIDI.

Main Operation

- **SHIFT** changes the function layer of the six knobs.
- **Short DICE** randomizes the sound while keeping the current performance mode.
- **SHIFT + short DICE** performs a deeper randomization.
- **Hold DICE for about 2 seconds** switches between Run mode and Touch mode.
- **Touch pads T1, T2, T3** reshape the system in Run mode and open the sound in Touch mode.

After randomization or a layer change, the knobs use pickup behavior: a knob must move away from its previous physical position before it takes control. This prevents sudden parameter jumps.

CV and MIDI

- **CV1** works as a trigger and clock input. Each rising edge strikes and nudges the synthesis core.
- **CV2** reacts to rising pulses as an additional trigger/clock source and also sends its level as MIDI CC 74.
- CV1 and CV2 pulses generate short MIDI notes 36 and 37 on MIDI channel 1.
- Incoming MIDI notes retune and strike the sound; note velocity controls the strength of the impact.
- Incoming MIDI Clock drives repeated strikes. Start and Continue also wake the engine.
- When no external clock is present, BlipOS generates MIDI Clock from Knob 1's internal speed setting.
- Incoming MIDI notes, CC, aftertouch, pitch bend, and realtime transport/clock messages are passed through.
- The twelve knob-layer values are available over MIDI CC: normal-layer controls use CC 20-25 and SHIFT-layer controls use CC 26-31.

LEDs

- The SHIFT and non-SHIFT LEDs show the active knob layer.
- The four lower LEDs display activity from the two internal shift registers.
- In Touch mode, their brightness follows the touch gate and audio activity.
- Short flashes confirm randomization and switching between Run and Touch modes.

Run Mode

Run mode leaves the BlipOS core continuously open. Two oscillators clock each other's shift-register states, while sample-and-hold feedback changes oscillator rates and separates the two resonant peaks. The interaction is deterministic but unstable enough to produce patterns that feel alive and difficult to repeat exactly.

Run Controls

Normal layer:

- **Knob 1:** Oscillator A rate / internal MIDI clock speed.
- **Knob 2:** Oscillator B pitch relationship.

- **Knob 3:** Resonant Peak 1 frequency.
- **Knob 4:** Resonant Peak 2 frequency.
- **Knob 5:** Peak spread, modulation depth, and excitation.
- **Knob 6:** Output gain and grit.

SHIFT layer:

- **Knob 1:** Rungler modulation sent to Oscillator A.
- **Knob 2:** Rungler modulation sent to Oscillator B.
- **Knob 3:** Sample-and-hold modulation sent to Oscillator A.
- **Knob 4:** Sample-and-hold modulation sent to Oscillator B.
- **Knob 5:** Shift-register source balance and resonant-peak modulation direction.
- **Knob 6:** Resonance, stereo spread, grit, and the relationship between the two oscillator rates.

Run Touch Pad Behavior

- **T1:** Pushes oscillator rate movement and makes the clock interaction more animated.
- **T2:** Increases rungler influence and pattern instability.
- **T3:** Emphasizes resonant-peak movement and ringing character.
- Longer holds deepen each pad's effect.

Touch Mode

Touch mode uses the same cross-coupled BlipOS engine, but closes its output until a touch pad, CV pulse, or MIDI note opens the gate. It is not a conventional three-voice mode: all three pads excite and reshape one shared stereo synthesis system.

- Touching any pad opens the sound.
- Each pad retains its own gesture role from Run mode.
- Longer holds gradually open the resonant system further, producing longer and brighter tones.
- CV1, CV2, and incoming MIDI notes can trigger the gate without touching a pad.
- The knob layout remains the same as in Run mode.

Performance Notes

- Knob 5 and SHIFT Knob 6 are the quickest route from sparse clicks to sustained resonant textures.
- High modulation and resonance settings can reorganize the pattern dramatically after a single trigger.
- In Touch mode, short taps behave like struck resonators, while longer holds reveal drones and feedback motion.
- BlipOS includes a silence-protection system that nudges the internal state away from dead zones while preserving the current character where possible.